慧谷
HUMANCHEM

THE SCIENCE

HUMAN NEW MATERIAL

BARRIER COATING FOR PAPER PACKAGING -FLUORINE-FREE, WATER AND OIL RESISTANT OVERALL SOLUTION









www.humanchem.com

About Humanchem

Guangzhou Human New Material Science and Technology Co.,Ltd.(Previously known as Guangzhou Human Chemicals Co.,Ltd.)was established in 1999, which is one of the first high-tech enterprises in Guangdong Province, specializing in R&D, manufacturing and sales of functional materials such as coatings, resins, adhesives. We have provided high-quality functional products and effective solutions for many industrial applications such as New Energy, Home Appliances, Electronics, Packaging, Digital Printing, etc.

Headquartered in Guangzhou, we have set up 2 plants respectively located in Guangzhou city and Qingyuan city,Guangdong province, covering a total area of more than 300,000+m^2. Now we have more than 700 employees, 200+ of whom are doctors, masters and senior engineers.

Laboratory area amounts to 10,000m^2, consisting of several R&D labs, verification platforms & materials analysis center, allowing Guangzhou Human New Material to continuously serve customers quickly and efficiently with innovative products. 6-8% of company's annual revenue has been invested in R&D.

Health, safety and environment integrated equipment helps reduce the impact of industrial production on environment.
Through more than 20 years of efforts and dedication Guangzhou Human New Material has received following honors:

- ◆National Manufacturing Single Champion Enterprise (Hydrophilic coating for A/C)
- ◆National high-tech enterprise
- ◆"Specialized in special new" medium-sized enterprises of Guangdong Province
- ◆Guangdong Provincial Enterprise Technology Center
- ◆Guangdong Engineering Technology Research and Development Center
- ◆Guangdong Science and Technology Award

We are expecting a greener Human New Material, forging ahead on the road of specialized and refined development.
Human New Material, Total solution for a better life.

FACTORY INTRODUCTION

Guangzhou

Guangzhou Human New Material Science and Technology Co., Ltd is the group headquarters.



Qingyuan

Human New Materials Qingyuan Factory was completed and put into operation in May 2018 as the group production

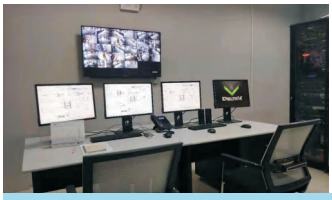


MANUFACTURING BASE





Manufacturing Base





Automatic Control System

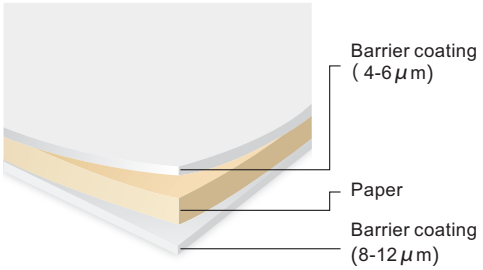
**Actively respond to the 14th Five-Year Plan,
positively promoting plastic reduction and VOC free in the packaging field**

Strive to become a major supplier of advanced organic materials in the sectors of electronics, display, new energy and other related fields

Waterborne Food-grade Low Temperature Heat Sealing Coating Non-PE Paper Cup

This series of coating is a waterborne heat-sealing coating, suitable for the surface coating of NON-PE paper cups, paper bowls, paper boxes and so on. It has features like excellent heat sealing, anti-blocking and water resistance.

- Low temperature heat sealing
- 60℃ Blocking resistance
- Low Cobb
- High kit value
- Fluorine-free







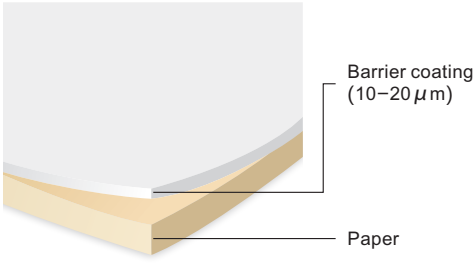
Items	Solid Content/%	T-4 Cup/S	Applications	Temperature Condition(range)	Cobb (8.5-9g/m²)	Kit (8.5-9g/m²)	Anti-blocking	Notes
HZ1320	35	20	Pre-coated paper for plastic-free cups	170-190℃*0.2s	＜10g/m2	Grade 10	Excellent	Suitable for medium and high speed cup making machine.
HZ1301	30	20	Pre-coated paper for plastic-free cups	170-190℃*0.2s	10g/m2	Grade 8	Excellent	Suitable for medium and high speed cup making machine.
HZ1400A	43	18	Pre-coated paper for plastic-free cups	210-230℃*0.2s	5g/m2	Grade 12	Fair	Suitable for medium and high speed cup making machine.

Waterborne Anti-dusting Coating & Waterborne Anti-static Coating

The waterborne anti-dusting coating is suitable for the surface treatment of industrial molded pulp. It imparts excellent anti-dusting properties and a superior smooth feel, ensuring full compliance with RoHS and REACH standards.

The waterborne anti-static coating is suitable for the surface treatment of industrial molded pulp trays. It imparts a stable surface resistance of 10^4–10^8 Ω, effectively eliminating static electricity, and ensures full compliance with RoHS and REACH standards.

- Anti-dusting, can pass the vibration abrasion test by 4H
- Superior smooth feel
- Excellent blocking resistance
- Compliant with RoHS and REACH







Items	Solid Content/%	T-4 Cup/S	Applications	Features	Process
HZ9400	40	60	Suitable for 3C package boxes, Molded pulp for industrial package	Matte、Superior smooth feel、Vibration abrasion resistance、Anti-dusting	Air spray and airless spray
HZ9600	52	70	Suitable for electronic component trays, molded pulp for industrial package	Provides static dissipation, with surface resistance achieving 10^4-10^8 Ω.	Air spray and airless spray

Fluorine-Free & Internal Sizing oil-resistant Agent

Fluorine-free waterborne polymer is applied to molded pulp products, it provides excellent and long-lasting water resistance (95 ± 5℃) and oil resistance (0-100℃).

- Fluorine-free, suitable for GB9685-2016
- Oil-resistant(0-100℃)
- Long-lasting performance



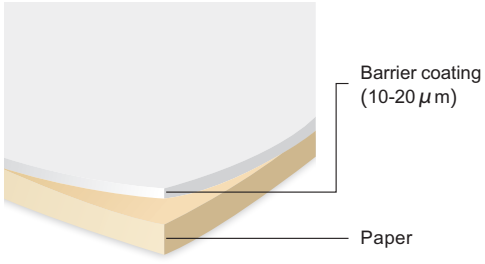


Items	Solid Content/%	Viscosity/ mPa.s	pH	Water solubility	Oil-Resistant (0-100℃)	Oil-Resistant (95±5℃)	Features	Applications
FY2507	17	150	5.0	Fully Soluble	√	√	Internal sizing; excellent water-resistant and oil-resistant; long-lasting	Molded pulp products

Waterbone Food-Grade Paper Barrier Coating – Molded Pulp

Waterbone food-grade paper barrier coating is suitable for both surface treatment of molded pulp and paperboard, which can achieve water-resistant, oil-resistant, anti-blocking, Fluorine-free, also can meet the requirements of GB9685 and GB4806.

- High-temperature oil resistant 150°C for 30 mins
- Hot-water resistant 100°C for 30mins
- Non-sticky rice
- Recyclable
- Fluorine-free



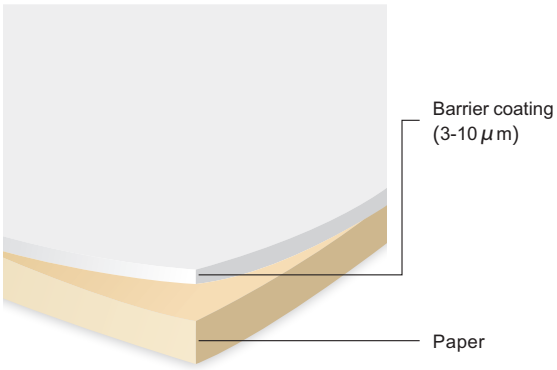



Items	Solid Content/%	T-4 Cup/S	Applications	Features	Process
HZ5300	37	20	Molded Pulp Food Tableware	Matte、Water resistant、Oil resistant、Good blocking resistance、Microwave heating、Non-sticky rice	Coating/Water=100/10 dilution, Air spray and airless spray.;
HZ5410A	43	15	Molded Pulp Food Tableware	Gloss、Water-resistant、Oil-resistant、Excellent anti-blocking、Microwave heating、Non-sticky rice、30% reduction in coating usage	Air spray and airless spray.
HZ5410B	43	40	Paper Food Container	High gloss, Water resistant、Oil resistant Blocking resistance	Anilox Roll、Flexography、Gravure Coating
HZ5306	36	13	Molded Pulp Food Tableware	High gloss, Water resistant、Oil resistant Blocking resistance	Suitable for paper knives, forks and spoons in dip-coating process
HZ5308	40	15	Molded pulp tableware for heating in food ovens.	Gloss、Oven heating、Microwave heating、Water-resistant、Oil-resistant、Excellent blocking resistance	Air spray and airless spray
HZ1320	30	20	Molded pulp tableware for PE heat sealing.	Heat sealing、High glossing、Water-resistant、Oil resistant、blocking resistance、Non-sticky rice	Air spray and airless spray

Waterborne Heat Sealing Coating Non-PE Paper Bags

This series of coating is a waterborne acrylic acid heat sealing coating, suitable for the surface coating of non-PE packaging paper bags, having excellent features like heat sealing, anti-blocking.

- Good heat sealing
- 60℃ Blocking resistance
- Fast-drying
- Fluorine-free





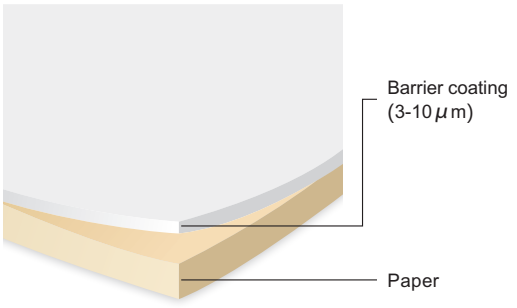


Items	Solid Content/%	T-4 Cup/S	Applications	Heat sealing temperature (Upper limit)	Anti-blocking	Notes
HZ1400	43	18	Heat sealing pre-coated paper for plastic-free paper bags	150℃*0.3s	Excellent	Suitable for coating machines
HZ2500N	50	45	Plastic-free high-strength heat sealing pre-coated paper	120℃*0.2s	Fair	Suitable for flexographic machines
LZG-240	35	25	Pre-coated paper for plastic-free paper bags	150℃*0.2s	Excellent	Suitable for coating and gravure machines

Waterborne Barrier Coating for Replacing Plastic Lamination

Waterborne barrier coating is suitable for kraft paper tape as a replacement for plastic layers and barrier coatings, having the features like excellent kit-resistant, barrier properties, good fastness to solvent-free silicone oil, anti-blocking.

- Oil-resistant、Water-resistant
- 60℃ Blocking resistance
- High kit value
- Fluorine-free







Items	Solid Content/%	T-4 Cup/S	Applications	Kit (4-5g/m^2)	Cobb	Anti-blocking	Notes
HZ8401	43	15	Pre-coated paper of plastic-free and oil-resistant. (Plastic-free paper tape, silicone oil paper)	Grade 9	< 7g/m2	Excellent	Suitable for coating machines

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